

Master Plan

Master Dashboard - Live Page Builder CRM - Quotations & Invoices - Go-Live

Full-stack implementation plan based on the completed deep audit,
the approved design mockup and the five locked decisions.

87-page website - No-code block builder - Lead-gen CRM with live chat & WhatsApp - Server-PDF
quotation pipeline - Vercel hosting.

Status	FOR APPROVAL - development starts after sign-off
Version	1.0 - 28 September 2026
Scope	Phases 0-6 - 13 weeks
Hosting	Vercel (site + admin + API) - Hostinger = domain only
Based on	AUDIT-REPORT-2026-09-28.md + design mockup + admin screenshots

WOODEX - Backend Master Plan

Version: 1.1 - **Date:** 2026-09-29 - **Status:** awaiting approval (phase gates) **Relationship:** extends MASTER-PLAN.md (Phases 0-6). Phase 0, 1, 2 are delivered; this plan governs the **backend/module build-out** that follows, in build order.

v1.1 adds: Connectivity Verification (B4), Site Control center, CRM Control Room (B7: dashboard + import/export + full tracking), live-chat agent console scope (B5.1), client database design (Appendix A), settings button/contrast + session fixes (done).

0. Current State (what is live vs planned)

Module	State
Dashboard home (KPIs, analytics, widgets)	? live
Page Builder + typed blocks (87 pages converted)	? live
Save draft -> Publish -> GitHub -> live	? live (tested end-to-end)
Enquiries (CRM list) + Page Builder	? live
Reports (range, KPIs, chart, CSV)	? live (fills after Supabase migration)
Settings buttons contrast + session-expiry sign-in bounce	? fixed this session
Pages management (add/import/status/SEO)	? B1
Live Chat (visitor + agent console), Header/Footer/Global Sections, Galleries, Image Optimization, Schema, WhatsApp module, Integrations + connectivity checks	? placeholders
CRM Control Room (dashboard, import/export, full tracking)	? B7
Runtime backup import/export	? B3

Pages vs Visual Builder (today): both open the same combined editor screen. **B1 splits them:** *Pages* = management of all 87 pages (list, add, import, status, SEO, duplicate, delete-with-redirect); *Visual Builder* = the section editor.

Decisions required (approval gate 0)

- **D6 - WhatsApp number.** Site currently uses **+92 322 4000768** (wa.me/923224000768, 230 occurrences). Requested number was **+92 322 4200768**. Confirm which is correct before any global replace (one digit differs: 4000768 vs 4200768).
- **D7 - WhatsApp scope.** Original D2 was "wa.me only". B2 proposes adding Meta Cloud API for **outbound notifications** (new enquiry -> WhatsApp alert to admin, quotation sent -> client ping). wa.me remains the customer-facing click-to-chat.
- **D8 - AI integrations.** B4 connectors: **Claude (Anthropic), Codex/OpenAI, Hermes** - used for content drafts, meta descriptions, reply suggestions. Confirm providers + that API keys are customer-supplied (stored server-side only).
- **D9 - Hosting connectivity.** For the connectivity verifier: Hostinger check = DNS lookup only (D1 keeps Hostinger DNS-only); confirmation not needed unless you also want a Hostinger API module.

B1 - Page Management System (87 pages) - *priority*

Goal: Pages becomes the control room for every page; Visual Builder is the editor.

Deliverables

1. **Pages list view** (view-pages, cms-pages?action=list v2)
 - All 87 pages: title, slug, type (service/location/insight/project/...), status (Published / Draft-changes-pending), SEO score (title+desc+canonical+OG present), sections count, last published (GitHub commit time), open-editor / view-live links.
 - Search box + status filter + type filter; sortable columns.
2. **Add Page wizard** (blueprint Section 4): Step 1 type -> Step 2 template donor -> Step 3 content (title/slug/hero placeholder) -> Step 4 SEO -> Step 5 Preview -> Publish. Backend: cms-pages?action=create extended with template presets.
3. **Page import system**
 - cms-pages?action=import&slug= - re-split a live page into blocks+meta (promotes tools/blocks-convert.py logic to a server module _page-split.mjs shared with the offline tool; identity validation required before overwrite).
 - Import external HTML: paste/upload HTML -> validate (main-element required, scripts same-origin only) -> register page.
 - "Re-sync from live" per page (discard local drafts, rebuild from current HTML).
4. **Page actions:** duplicate (new slug + donor copy), archive (hide from sitemap, keep file), delete with redirect (exists - surfaced in list UI).
5. **Visual Builder entry:** editor deep-link ?page=/slug/ opens straight into the builder; sidebar Visual Builder = "resume last edited".

API

GET list|get - POST save|publish|create|import|resync|duplicate|archive|delete

Acceptance gate (B1)

- 87 pages listed with correct status/SEO columns; search+filters work.
- Wizard creates a page end-to-end (draft -> preview -> publish -> live URL).
- Import of a live page reproduces it byte-identical (identity check) before any edit.
- Duplicate produces an independent page; delete inserts redirect row.

B2 - WhatsApp API (number per D6/D7)

Deliverables

1. **Settings -> WhatsApp:** number, greeting text, off-hours message, admin notify toggle. Single source of truth (blueprint Section 23) - header, footer, floating button, contact page all read it (replace the 230 hardcoded occurrences with assets/js/site-config.js values, published via cms-settings).
2. **wa.me click-to-chat** everywhere (already present; re-pointed from settings).
3. **Cloud API outbound (if D7 approved):** netlify/functions/whatsapp-send.mjs - server-side Graph API call (token env WHATSAPP_TOKEN, phone id env), used by: new-enquiry admin alert, quotation-approved notification. Template messages only (Meta policy), rate-limit + idempotency key.
4. **Inbound (phase 2 of B2):** webhook endpoint whatsapp-webhook.mjs -> creates/updates enquiry in CRM with source=whatsapp. Verify HMAC (X-Hub-Signature-256).

Acceptance gate (B2)

- Number editable in Settings -> updates all site touchpoints after publish.
- Test alert: create enquiry -> admin WhatsApp received (or toggle off gracefully).
- Webhook (if enabled) creates a CRM row; invalid signatures rejected.

B3 - Runtime Backup / Import / Export

Goal: never depend on manual exports; any point-in-time restore.

Deliverables

- Export** (cms-backup?action=export&scope=all|cms|content|media)
 - JSON dump of all CMS tables (Supabase REST), content/pages/ (**drafts**), **vercel.json redirects**, **settings row**; **media** manifest (**URLs+metadata**).
 - Streams as single .json (zip when media binary included).
- Scheduled GitHub backup:** nightly cms-backup?action=snapshot committing backups/YYYY-MM-DD.json to a backups branch (keeps 14 days, oldest pruned).
- Import / restore** (cms-backup?action=import)
 - Upload JSON -> schema validation -> **dry-run diff preview** -> confirmed apply; every import writes an audit-trail entry.
- UI: improve existing **Backups** view - snapshot list, Download, Restore, Export-now, Last-backup status on System Health.

Acceptance gate (B3)

- Export downloads a valid JSON that passes import dry-run with 0 errors.
- Snapshot branch shows dated commits; retention prunes at 14.
- Restore of a dry-run-tested file reproduces CMS rows (tested on a sandbox page).

B4 - Integrations Hub + Connectivity Verification (Settings -> Integrations)

Goal: one place to connect AI + business services **and to see exactly what is connected, what is broken, and what is missing**.

B4.0 - Connectivity Verification System (new in v1.1)

"Where am I missing a connection?" - a status matrix, one row per service:

Service	Check performed	Needs
GitHub	authenticated API ping + branch exists + write probe (issue a test blob)	CMS_GITHUB_TOKEN etc.
Supabase	REST ping + cms_users/settings tables exist + migration fingerprint	SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY
Vercel	deployment status via API + domain bound	VERCEL_TOKEN, project id
Hostinger	DNS resolution of woodex.com.pk -> expected target (A/CNAME)	none (D1: DNS only)
Resend / WhatsApp / AI keys	provider ping (/v1/messages, /v1/models, ...)	per-provider key

- GET cms-connect?action=check runs all probes in parallel -> per-service {ok, latency, detail, missing_env:[...], fix_hint}.
- UI: green/amber/red cards + **"Fix"** deep-link (which Vercel env var, which

migration step) - turns checklist errors into one-click diagnostics.

- Runs automatically on System Health load and on demand ("Re-check connections").

B4.1 - Integrations model

- Table integrations: provider, name, api_key_enc, config jsonb, enabled, updated_by, updated_at. Key write-only from UI (masked ----1234 display), service-role only on server.

B4.2 - Providers (initial)

- **Claude (Anthropic) - Codex (OpenAI) - Hermes** - content drafts, FAQ generation, reply suggestions (adapter _ai.mjs, provider selected from enabled integrations; output lands in editor fields, never auto-published).
- **Resend (D5)** - quotation/invoice email.
- **GitHub / Vercel / Supabase / Hostinger** - read-only status (B4.0).

B4.3 - Site Control center (Settings upgraded)

- One screen: contact/social/estimator rates (today) + WhatsApp (B2) + integrations & connectivity (B4.0) + publish switches (maintenance mode, cookie notice, chat widget on/off) - everything that controls the website, each with last-saved time + Publish-to-site action (existing pipeline).

B4.4 - Webhooks out

- Generic webhook URL + event list (enquiry.created, quotation.sent, page.published) -> HMAC-signed POST.

Acceptance gate (B4)

- Connectivity matrix renders all services with correct green/red + missing-env hints.
- Save key -> test -> green; key never visible again in any response.
- One AI surface (meta description) generates into the Pages SEO form end-to-end.
- Webhook fires on enquiry.created with valid signature (verified by test receiver).
- Site Control publishes a switch (e.g., chat widget off) to the live site.

B5 - Blueprint modules (build order)

#	Module	Notes
B5.1	Live Chat - full agent console	Visitor side: widget on every public page, offline -> WhatsApp fallback. Agent side (dashboard): conversation list w/ unread badges, thread view, agent replies (live via Supabase Realtime), canned replies, visitor context (page, location, history), sound+desktop notification on new message, after-hours auto-reply, transcript saved to CRM timeline. Tables: chat_channels/messages/agents (migration add-on).
B5.2	Header & Navigation builder	Controls per blueprint Section 8; writes header include used by publish pipeline.
B5.3	Footer editor	Same pattern; global footer section.
B5.4	Global Sections	Reusable CTA/banner sections referenced by pages (block type global_ref, resolved at publish).
B5.5	Galleries + Image Optimization	Media folders/albums; sharp/AVIF pipeline via cms-media extension + nightly optimize job.
B5.6	Schema manager	Per-page JSON-LD types (FAQ, Service, LocalBusiness, Article) rendered into <head> at publish; validation on save.

#	Module	Notes
B5.7	Activity	Activity feed from audit_trail (Section 28), filtered.

Each module = its own approval gate (build -> preview -> "done ?" -> next).

B6 - Backend hardening (continuous, parallel)

1. **API envelope:** consistent {ok|error, data} + machine error codes.
2. **Rate limiting:** IP+route buckets for public endpoints.
3. **Audit coverage:** every mutation writes audit_trail.
4. **Secrets hygiene:** keys in Vercel env or integrations table - never repo; rotation checklist at go-live (exposed setup credentials included).
5. **CI smoke suite:** tools/smoke.mjs (syntax, 86-page render identity, block identity, nav/TITLE integrity) + GitHub Action on push.
6. **Health enrichment:** System Health gains CMS API, Media, Publishing, Migrations rows (Section 27) + B4.0 connectivity matrix.
7. **Sitemap automation:** regenerated on every publish.

B7 - CRM Control Room - *new in v1.1*

Goal: one screen to run the whole client lifecycle: lead -> estimation -> quotation -> invoice -> payment, with full control (import/export, tracking, filters).

B7.1 - CRM Dashboard

- Funnel KPIs: Enquiries -> Qualified Leads -> Estimations -> Quotations -> Invoices -> Paid (counts + value per stage, conversion % between stages).
- Pipeline board (drag or status menu), value-weighted (PKR), per-user ownership.
- Follow-ups due today, visits this week, overdue invoices - action-required rail.
- Cohort strip: response time avg, win rate, avg quotation -> invoice lag.

B7.2 - CRM Import / Export

- **Export:** CSV/JSON for leads, clients, quotations, invoices (filter by date, status, source); one-click "full CRM backup" routed through B3 snapshot.
- **Import:** CSV upload per entity -> column mapper (detect headers) -> validation preview (bad rows listed with reasons) -> dry-run diff -> commit; duplicates matched on email/phone; every import logged to audit_trail.
- Round-trip guarantee: export -> import -> identical rows (tested in gate).

B7.3 - Tracking system (linked lifecycle)

- **Lead tracking:** status timeline (New -> Contacted -> Qualified -> Lost/Won), stage history with who/when, source attribution (form, WhatsApp, chat, call), notes + attachments, SLA timer (first response).
- **Estimation tracking:** estimator submissions linked to leads; saved configurations re-price with current rate card; send-for-quote button.
- **Quotation tracking:** statuses (Draft -> Sent -> Viewed -> Accepted -> Rejected -> Expired), version history (revisions keep old PDFs), reminder schedule, link-view tracking (opened at).

- **Invoice tracking:** Draft -> Sent -> Partial -> Paid -> Overdue; payment log (date, method, reference); auto-overdue flagging; balance on client card.
- Every stage change writes CRM timeline entry; all four entities share client_id + lead_id joins (Appendix A).

B7.4 - Client database (Appendix A) surfaced in UI

- Client card: contact, addresses, linked leads/quotations/invoices, communications (WhatsApp/chat/email log), documents, lifetime value.

Acceptance gate (B7)

- Import a 50-row lead CSV (incl. 3 intentionally bad rows -> clear errors) -> commit -> export -> row count and values match.
- Advance a lead through all 5 stages -> timeline shows 5 entries with authors.
- Quotation: send -> mark viewed -> accepted -> invoice auto-created pre-filled -> mark paid -> client card lifetime value updates.
- Dashboard funnel numbers reconcile with list views (same filters).

Suggested build order & gates

```
D6/D7/D8 confirm --? B1 Pages management --? B3 Backup/Export --? B4 Integrations
| | + B4.0 connectivity
| +--? B7 CRM Control Room ?-- (needs B1 patterns, B3 safety)
+--? B2 WhatsApp (D7) ^
B5.1 Live Chat (parallel with B1 - realtime, independent)
B5.2-B5.7 after B1 (all touch publish pipeline)
B6 continuous throughout
```

- **Gate rule (unchanged):** each B-phase ends with your test on the preview + "done ?" before the next starts. Deployment to Vercel stays the final step after your overall approval.

Appendix A - Client database design (v1.1)

```

clients
id uuid pk - company_name - contact_name - email - phone - whatsapp
type (individual|company) - billing_address - notes - source
status (active|archived) - lifetime_value numeric - created_at - updated_at

leads (B7.3)
id - client_id fk? - enquiry_id fk? - title - status (new|contacted|
qualified|won|lost) - source (form|whatsapp|chat|call|referral) -
value_estimate numeric - owner (username) - next_followup_at -
first_response_at - created_at

estimator_submissions
id - lead_id fk? - service - tier - area - config jsonb -
price_quote numeric - status (received|quoted|converted)

quotations
id - lead_id fk? - client_id fk? - number (WX-Q-YY-NNN) -
status (draft|sent|viewed|accepted|rejected|expired) - items jsonb -
subtotal - tax - discount - total PKR - pdf_path - send_at -
view_at - respond_at - valid_until - template_id - version int

invoices
id - quotation_id fk? - client_id fk? - number (WX-I-YY-NNN) -
status (draft|sent|partial|paid|overdue) - items jsonb - total -
paid_amount - due_date - payments jsonb [{date, amount, method, ref}] -
pdf_path - sent_at - paid_at

crm_timeline (single feed for all entities)
id - entity (lead|quotation|invoice|client|chat|enquiry) - entity_id -
event (created|stage_changed|sent|viewed|reminder|payment|note|...) -
from_status - to_status - actor - message - created_at

communications_log
id - client_id - channel (whatsapp|chat|email|call) - direction -
summary - ref_url - at - actor

chat_channels / chat_messages (B5.1)
channels: id - visitor_id - page - status (open|waiting|closed) -
assigned_to - unread_admin - created_at - last_msg_at
messages: id - channel_id - from (visitor|agent) - body -
agent_name - read_at - created_at

```

Indexes: leads(status, owner) - quotations(status) - invoices(status, due_date) - crm_timeline(entity, entity_id) - clients(phone)/(email) (import matching).

Go-live linkage (from MASTER-PLAN)

Renames/keeps: media metadata columns, project publishing columns, cms_users, role testing, test publish, settings publish, desktop/mobile dashboard test, public page tests, **credential rotation**, final acceptance - executed with B3 (backup safety net in place first) immediately before cutover.

Changelog

- **v1.0** - initial: B1 page management, B2 WhatsApp, B3 backup, B4 integrations, B5 modules, B6 hardening; decisions D6-D8.

- **v1.1** - + B4.0 connectivity verification (GitHub/Supabase/Vercel/Hostinger

missing-connection detector), B4.3 Site Control center, B7 CRM Control Room (dashboard, CSV import/export, lead/estimation/quotation/invoice tracking), B5.1 expanded to full agent console, Appendix A client DB design, D9. Recorded fixes: settings button contrast pass + 401 sign-in bounce.